

Hardware Interface: Dedicated UART2 Interface on ESP32 | **RX Pin:** GPIO 16 | **TX Pin:** GPIO 17 | **Electrical:** 3.3V LVTTTL / RS-232 Transceiver | **Framing:** ASCII CRLF (\r\n) Delimited

1. PRIMARY REAL-TIME TELEMETRY STREAM (\$PCU)

Transmitted continuously by the solar inverter controller (recommended interval: **1000ms – 2000ms**). Supports standard 15-parameter and extended 19-parameter formats.

```
$PCU,<mains>,<solar>,<batt>,<acout>,<load>,<chg>,<disch>,<dc>,<heat>,<onflag>,<solarstate>,<chargerstate>,<dcok>,<sharemode>,<batgravity>[,<switchstate>,<dcboostmode>,<feedmode>,<error_code>]\r\n
```

#	Field Name	Type	Unit	Valid Range	Functional Description & Inverter Engineering Role
1	mains	Float	VAC	0.0 – 450.0	AC Utility Mains Grid RMS Voltage. Triggers grid icon, pass-through, and grid chargers.
2	solar	Float	VDC	0.0 – 250.0	Solar PV array DC terminal voltage (MPPT operating voltage or open-circuit Voc).
3	batt	Float	VDC	0.0 – 100.0	Battery bank terminal voltage. Drives state-of-charge (SoC) gauges and bars.
4	acout	Float	VAC	0.0 – 450.0	Inverter output voltage delivered to connected household/industrial AC loads.
5	load	Float	%	0.0 – 250.0	Active inverter load demand as a percentage of rated capacity (100% = Full Rating).
6	chg	Float	A	0.0 – 200.0	Battery charging current generated from Solar PV and/or Grid Mains charger stages.
7	disch	Float	A	0.0 – 200.0	Battery discharge current supplied to inverter H-bridge during power outages.
8	dc	Float	VDC	0.0 – 600.0	Internal high-voltage DC link bus voltage before the sinusoidal bridge.
9	heat	Float	°C	-20.0 – 150.0	Power stage MOSFET/IGBT heatsink thermal sensor reading.
10	onflag	Integer	Binary	0 or 1	Master inverter system state: 0 = Standby / Inactive, 1 = Inverter System Active.
11	solarstate	Integer	Enum	0 or 1	Solar PV array status: 0 = Solar Inactive (Night), 1 = Solar Active (Generating).
12	chargerstate	Integer	Enum	0 – 3	0 = Off, 1 = AC Grid Charger, 2 = Solar MPPT Charger, 3 = Shared / Hybrid Charger.
13	dcok	Integer	Binary	0 or 1	Internal DC Boost Bus health monitor: 0 = Bus Fault, 1 = Bus Healthy.

#	Field Name	Type	Unit	Valid Range	Functional Description & Inverter Engineering Role
14	sharemode	Integer	Binary	0 or 1	Power distribution mode: 0 = Standalone Invert, 1 = Smart Hybrid Grid Sharing.
15	batgravity	Integer	Index	0 – 10	Electrolyte specific gravity / battery chemical condition index.
16*	switchstate	Integer	Binary	0 or 1	Inverter power bridge state: 0 = Bridge Off, 1 = Bridge Active Switching.
17*	dcboostmode	Integer	Binary	0 or 1	DC Booster state: 0 = Direct / Bypass, 1 = Active Boost Stage.
18*	feedmode	Integer	Enum	0 – 3	0 = Auto Physics, 1 = Solar Priority, 2 = Battery Invert, 3 = Grid Pass.
19*	error_code	Integer	Enum	0 – 40	System fault code: 0 = Normal Operation. Any code > 0 triggers Protection Screen.

* Parameters 16 through 19 are optional extended parameters. If omitted, firmware auto-derives states safely.

Sample \$PCU Data Frames

- **Optimal Solar Daytime Operation:**
\$PCU,230.0,78.5,27.2,230.0,42.0,18.5,0.0,380.0,41.5,1,1,2,1,0,0,1,1,1,0\r\n
- **Night-time Battery Inversion (Mains Grid Outage):**
\$PCU,0.0,0.0,24.6,230.0,55.0,0.0,22.4,360.0,48.0,1,0,0,1,0,0,1,1,2,0\r\n
- **Utility Grid Charging (Rainy / Night):**
\$PCU,228.0,0.0,25.8,228.0,28.0,15.0,0.0,320.0,38.0,1,0,1,1,0,0,1,0,0,0\r\n

2. CALIBRATION & SAFETY LIMITS STREAM (\$LIMITS / \$CFG)

Transmitted at system startup or when battery/grid thresholds are configured. The HMI display controller stores these calibration limits to evaluate trip conditions and display reference lines.

```
$LIMITS,<setbatful>,<setbatwrn>,<setbatlo>,<setbatrst>,<mainslow>,<mainshi>,<hiheat>,<solmax>,<solmin>,<dcmax>\r\n
```

#	Field Name	Type	Unit	Default	Calibration Range	Engineering Definition & Role
1	setbatful	Float	VDC	28.8 V	24.0 – 32.0 V	Battery Absorption / Float target voltage threshold.
2	setbatwrn	Float	VDC	22.0 V	21.0 – 24.0 V	Battery Low Warning advisory threshold.
3	setbatlo	Float	VDC	21.0 V	19.0 – 22.0 V	Battery Deep Discharge Cutoff trip limit (protects cells).
4	setbatrst	Float	VDC	24.5 V	23.0 – 26.0 V	Battery Cutoff Reset / Reconnect voltage threshold.
5	mainslow	Float	VAC	185 VAC	150 – 200 VAC	AC Utility Grid Low Cutoff Voltage. Below this, mains trips.
6	mainshi	Float	VAC	265 VAC	240 – 290 VAC	AC Utility Grid High Surge Cutoff Voltage.
7	hiheat	Float	°C	85.0 °C	65.0 – 105.0 °C	Power stage thermal shutdown trip limit.
8	solmax	Float	VDC	115.0 V	80.0 – 220.0 V	Solar PV array maximum open-circuit voltage (\$V_{oc}\$).
9	solmin	Float	VDC	30.0 V	15.0 – 60.0 V	Solar PV minimum MPPT wake-up threshold.
10	dcmax	Float	VDC	410.0 V	350.0 – 550.0 V	Internal DC link bus maximum overvoltage trip threshold.

* Auto-Scaling Guard: If integer representations are sent (e.g. 288 for 28.8V), the parser automatically divides by 10.

Sample \$LIMITS String

```
$LIMITS,28.8,22.0,21.0,24.5,185.0,265.0,85.0,115.0,30.0,410.0\r\n
```

3. DEDICATED PROTECTION & FAULT CONTROL (\$FAULT / \$ERR)

When any fault code > 0 is received, the HMI carousel immediately freezes and locks onto the **Dedicated Theme-Independent Alert Red Protection Screen**. Sending 0 clears the fault.

```
$FAULT,<code>\r\n      (e.g., $FAULT,4 to trip on Inverter Overload)
```

Code	Severity & Status	Screen Display Title (Bold 20pt)	Offending Value & Limit Rendered on Screen
0	FAULT CLEARED	Resumes normal carousel	Restores active dashboard page automatically.
1	CRITICAL HARD TRIP	OUTPUT SHORT CIRCUIT	SHORT DETECTED Limit: OUTPUT IMPEDANCE VIOLATION

Code	Severity & Status	Screen Display Title (Bold 20pt)	Offending Value & Limit Rendered on Screen
2	PROTECTION TRIP	AC MAINS FEED LOSS	<mains> VAC Limit: NORMAL LIMIT: 185V - 265V
3	THERMAL SHUTDOWN	HEATSINK OVERTEMPERATURE	<heat> °C Limit: MAX SAFETY LIMIT: 85 °C
4	CAPACITY TRIP	INVERTER OVERLOAD	<load> % LOAD Limit: MAX RATED CAPACITY: 100 %
5	DC BUS TRIP	DC BUS UNDERVOLTAGE	<dc> VDC Limit: MINIMUM BUS LIMIT: 280 V
6	INRUSH CURRENT	OUTPUT CURRENT OVERLOAD	<load> % PEAK Limit: CYCLE-BY-CYCLE CURRENT LIMIT
7	SOLAR OVERVOLT	SOLAR PV OVERVOLTAGE	<solar> VDC Limit: MAX ARRAY LIMIT: 115 V
8	DC BUS OVERVOLT	DC BUS OVERVOLTAGE	<dc> VDC Limit: MAXIMUM DC CEILING: 410 V
9	HARDWARE FAULT	DC BOOST STAGE FAILURE	<dc> VDC Limit: BOOST CONVERTER REGULATION FAULT
20	SYSTEM ADVISORY	LOAD CAPACITY WARNING	<load> % LOAD Limit: WARNING THRESHOLD: 85 %
30	SYSTEM ADVISORY	BATTERY VOLTAGE LOW	<batt> VDC Limit: WARNING THRESHOLD: 22.0 V
40	BATTERY SHUTDOWN	BATTERY CUTOFF TRIP	<batt> VDC Limit: CUTOFF THRESHOLD: 21.0 V

4. MANUAL PAGE NAVIGATION COMMAND (\$PAGE)

Directs the display controller to immediately present a specific dashboard screen:

```
$PAGE,<0-5>\r\n      (e.g., $PAGE,1 to switch to Solar PV & Charger Page)
```

Page #	Screen Title	Primary Widgets & Metrics Displayed
0	ENERGY FLOW HUB	Live animated system power flow: Solar, Grid Mains, Battery bank, Inverter AC load.
1	SOLAR PV & CHARGER	Solar array voltage, charging current, MPPT active status badge, solar mode.
2	BATTERY & INVERTER LOAD	Battery terminal voltage, battery discharge Amps, active load percentage bar.
3	OPERATIONAL STATUS	Grid synchronism status, DC boost bus voltage, heatsink thermal sensor reading.
4	HARVEST & PRODUCTION	Solar energy yield (kWh counters) and historical 8-hour live line chart.
5	OEM VENDOR & SUPPORT	Hardware model, firmware release build, technical support contacts, custom OEM logo.